

TASK 7: Get Basic Sales Summary from a Tiny SQLite Database using Python

Objective: Use SQL inside Python to pull simple sales info (like total quantity sold, total revenue), and display it using basic print statements and a simple bar chart.

Tools:

- Python (with sqlite3, pandas, matplotlib)
- SQLite (built into Python — no setup!)
- Jupyter Notebook or a .py file

Dataset:

Create a small SQLite database file (sales_data.db) with just one sales table

Deliverables:

Python script (or notebook) that:

- Connects to sales_data.db
- Runs 1–2 SQL queries
- Displays output using print and basic matplotlib bar chart

Hints / Mini Guide:

- Load SQLite database: `import sqlite3 conn = sqlite3.connect("sales_data.db")`
- Run basic SQL: `query = "SELECT product, SUM(quantity) AS total_qty, SUM(quantity * price) AS revenue FROM sales GROUP BY product"`
- Load into pandas: `import pandas as pd df = pd.read_sql_query(query, conn)`
- Print results: `print(df)`
- Plot simple bar chart: `df.plot(kind='bar', x='product', y='revenue')`
- Save chart if needed: `plt.savefig("sales_chart.png")`

Outcome of Task:

By completing this task, you will:

- Learn how to write basic SQL queries
- Practice importing SQL data into Python
- Perform simple data summaries
- Create your first sales chart

Interview Questions:

- How did you connect Python to a database?
- What SQL query did you run?
- What does GROUP BY do?
- How did you calculate revenue?
- How did you visualize the result?
- What does pandas do in your code?
- What's the benefit of using SQL inside Python?
- Could you run the same SQL query directly in DB Browser for SQLite?

Task Submission Guidelines

- **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at

- **Self-Research Allowed:**

- You are DebugfreetoYourself:explore, Google, or refer to tutorials to understand concepts and complete the task effectively. Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

